How-To Library

Informatica

Optimizing Performance of an SAP Table Data Extraction Mapping

© Copyright Informatica LLC 2020, 2021. Informatica, the Informatica logo, and PowerExchange are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html.

Abstract

This article describes the parameters that you can tune to optimize the performance of an SAP Table Data Extraction mapping from Informatica® Developer.

Supported Versions

- PowerExchange for SAP NetWeaver 10.4.1

Table of Contents

Overview. 2
Performance Tuning Parameters. 2
Heap Size. 2
Packet Size. 3
Partitioning. 3

Overview

Performance tuning is an iterative process where you analyze the performance, use guidelines to estimate and define parameters that impact the performance, monitor, and adjust the results as required. Sizing is a process where you analyze the data set and use the recommended hardware to improve the performance.

This document describes the parameters that you can tune to optimize the performance of an SAP Table Data Extraction mapping from Informatica Developer.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of an SAP Table Data Extraction mapping might vary based on individual environments and other parameters even when you use the same data.

Performance Tuning Parameters

You can optimize the performance of an SAP Table Extraction mapping by tuning the following parameters:

- Heap size
- Transfer packet size
- Partitioning

Heap Size

To successfully run an SAP Table Extraction mapping to read data from SAP tables through the streaming mode, you must configure the Java heap size for the Data Integration Service.

Informatica recommends a minimum heap size of 1024 MB or higher when reading data from SAP tables using an SAP table extraction mapping in streaming mode.

Perform the following tasks to configure the Java heap size:

1. In the Administrator Console, navigate to the Data Integration Service for which you want to change the Java heap size.
2. Click the **Processes** tab.

3. Click **Custom Properties**.

   The Edit Custom Properties dialog box appears.

4. Click **New** to add a new custom property.
5. Specify the property name and value.

   The following table lists the property names and sample values:

Property Name	Property Value	Sample value
JVMOption1	-Xmx<memory_size> to set the maximum heap size	-Xmx1024m

6. Specify the maximum heap size based on the packet size value that you want to process.
7. Restart the Data Integration Service.

Packet Size

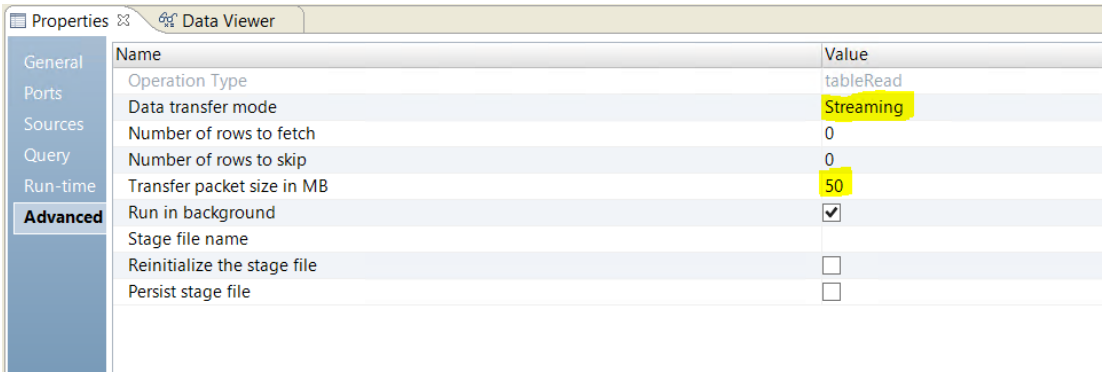
When you use the HTTP streaming data transfer mode in an SAP table data object read operation and run a mapping to read data from SAP tables, you can tune the HTTP transfer packet size to increase the throughput. The specified packet size and row size determine the number of rows that the Integration Service reads in a single packet.

Tune the packet size according to the network bandwidth, memory, and CPU resources available on the machine that runs the Data Integration Service. The maximum transfer packet size value must not exceed 999999. Default is 10 MB.

A higher packet size consumes more CPU and memory compared to the default value. If you increase the packet size, increase the heap size accordingly to improve the throughput.

You can configure the transfer packet size in the **SAP Table Data Object Read Operation** advanced properties. Select **Streaming** as the **Data Transfer Mode**, and then specify the transfer packet size in MB.

The following image shows the configured transfer packet size in MB for an SAP table data object read operation in the streaming data mode:



Partitioning

When you configure an SAP Table Data Extraction mapping to read data from SAP tables, you can configure partitioning to optimize the mapping performance at run time. Informatica recommends that you configure multiple partitions. You can define the partition type as key range partitioning.

The Data Integration Service distributes rows of data based on a port or set of ports that you define as the partition key. You can define a range of values for each port. The Data Integration Service uses the key and ranges to send rows to the appropriate partition.

When you configure partitioning, Informatica recommends that you set a minimum heap size of 1024 MB. You might have to increase the heap size as the number of partitions increase to optimize the performance.

For more information about partitioning, see the *PowerExchange for SAP NetWeaver User Guide*.

Authors

Gurkirat Singh

Dimple Rai